# THE W-LOCALIZATION OF AN AFFINE SCHEME IS IND-ZARISKI, FAITHFULLY FLAT AND HAS THE EXPECTED CLOSED POINTS AND SPECIALIZATIONS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the derivation of faithful flatness from surjectivity, the description of the closed points through the closures of the images in the partial w-localizations, the two-point separation argument proving uniqueness of specializations into Z, and the reducedness computation.

*Editorial note.* The label lemma-make-w-local names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Editorial note.* Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-make-w-local –start 843 –end 900 –local-id 3578 returned RESULT=OK (L3578 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

*Editorial note.* The proof refers to 'the reader verifies' for the elementary nilpotent computation that finishes part (4); that computation is a one-line verification of the displayed description of the Z_E rather than an unproved mathematical step, so the statement is counted with the phrase recorded here verbatim.

*Editorial note.* The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

*Allowed standard prerequisites (editorial).*

- Faithful flatness from surjectivity of the spectrum map. Algebra, the lemma that a ring map is faithfully flat if the induced map on spectra is surjective and the map is flat. Supplies the faithful flatness of A -> A_w once surjectivity on spectra is known.
- Structure of the partial w-localizations Z_E. Same chapter, the construction of the Z_E as disjoint unions of locally closed subschemes isomorphic to spectra of localizations of quotients. Supplies the description of Z_E used in the proof that the Z_E have no nilpotents and in the bijectivity Z -> X.
- Quasi-finiteness and open images for etale maps. Algebra, the lemmas on the local structure of the partial w-localizations and the finiteness of the primes lying over a maximal ideal. Supplies the identification of the closed points of the limit and the absence of specializations between the points of Z.

*Date*: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 00HQ: sources/stacks-chow/tags.txt, source lines 652–652. Prerequisite pointer invoked at proof line(s) 862; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 096A: sources/stacks-proetale/pinned-original-source.tex, source lines 405–405. Prerequisite pointer invoked at proof line(s) 854; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $X = \mathrm{Spec}(A)$ be an affine scheme. With $A \to A_w$, $X_w = \mathrm{Spec}(A_w)$, and $Z \subset X_w$ as above.*

1. *$A \to A_w$ is ind-Zariski and faithfully flat,*
2. *$X_w \to X$ induces a bijection $Z \to X$,*
3. *$Z$ is the set of closed points of $X_w$,*
4. *$Z$ is a reduced scheme, and*
5. *every point of $X_w$ specializes to a unique point of $Z$.*

*In particular, $X_w$ is w-local (Definition 096A).*

*Proof.* The map $A \to A_w$ is ind-Zariski by construction. For every $E$ the morphism $Z_E \to X$ is a bijection, hence (2). As $Z \subset X_w$ we conclude $X_w \to X$ is surjective and $A \to A_w$ is faithfully flat by Algebra, Lemma 00HQ. This proves (1).

Suppose that $y \in X_w$, $y \notin Z$. Then there exists an $E$ such that the image of $y$ in $X_E$ is not contained in $Z_E$. Then for all $E \subset E'$ also $y$ maps to an element of $X_{E'}$ not contained in $Z_{E'}$. Let $T_{E'} \subset X_{E'}$ be the reduced closed subscheme which is the closure of the image of $y$. It is clear that $T = \lim_{E \subset E'} T_{E'}$ is the closure of $y$ in $X_w$. For every $E \subset E'$ the scheme $T_{E'} \cap Z_{E'}$ is nonempty by construction of $X_{E'}$. Hence $\lim T_{E'} \cap Z_{E'}$ is nonempty and we conclude that $T \cap Z$ is nonempty. Thus $y$ is not a closed point. It follows that every closed point of $X_w$ is in $Z$.

Suppose that $y \in X_w$ specializes to $z, z' \in Z$. We will show that $z = z'$ which will finish the proof of (3) and will imply (5). Let $x, x' \in X$ be the images of $z$ and $z'$. Since $Z \to X$ is bijective it suffices to show that $x = x'$. If $x \neq x'$, then there exists an $f \in A$ such that $x \in D(f)$ and $x' \in V(f)$ (or vice versa). Set $E = \{f\}$ so that

$$X_E = \mathrm{Spec}(A_f) \amalg \mathrm{Spec}(A^\sim_{V(f)})$$

Then we see that $z$ and $z'$ map $x_E$ and $x'_E$ which are in different parts of the given decomposition of $X_E$ above. But then it impossible for $x_E$ and $x'_E$ to be specializations of a common point. This is the desired contradiction.

Recall that given a finite subset $E \subset A$ we have $Z_E$ is a disjoint union of the locally closed subschemes $Z(E', E'')$ each isomorphic to the spectrum of $(A/I)_f$ where $I$ is the ideal generated by $E''$ and $f$ the product of the elements of $E'$. Any nilpotent element $b$ of $(A/I)_f$ is the class of $g/f^n$ for some $g \in A$. Then setting $E' = E \cup \{g\}$ the reader verifies that $b$ is pulls back to zero under the transition map $Z_{E'} \to Z_E$ of the system. This proves (4). $\square$